For example the trend value for the year 1951 = 350.03 – 71.17 = 278.86

The value for the year 2011 = 634.7 + 71.17 = 705.87

The trend values have been calculated by successively subtracting and adding the increase in trend for previous and following years respectively.

Example 7.4

Find the trend values by semi-average method for the following data.

Year	1965	1966	1967	1968	1969	1970	1971	1972
Production of bleaching powder (in tonnes)	7.4	10.8	9.2	10.5	15.5	13.7	16.7	15

Solution:

Trend values using semi averages method

Year	Production of bleaching powder	4 year semi-total	4 year semi-average	Trend
1965	7.4	37.9	9.475	7.315
1966	10.8			8.755
1967	9.2			10.195
1968	10.5			11.635
1969	15.5	60.9	15.225	13.075
1970	13.7			14.515
1971	16.7			15.955
1972	15			17.395

Difference between the years = 1970.5 – 1966.5 = 4

Difference between the semi-averages = 15.225 – 9.475 = 5.75

Increase in trend = $\frac{5.75}{4} = 1.44$

Half yearly increase in trend = $\frac{1.44}{2} = 0.72$

The trend value for 1967 = 9.475 + 0.72 = 10.195

The trend value for 1968 = 9.475 + 3 * 0.72 = 11.635

Similarly the trend values for the other years can be calculated.

7.3.3 Moving Averages Method

Moving averages is a series of arithmetic means of variate values of a sequence. This is another way of drawing a smooth curve for a time series data.

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